

Khandaker Abrar Nadib

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RESEARCH INTERESTS

- Data Visualization
- Human-Computer Interaction
- Human-Centered AI, AI-Assisted Visualization Systems

PUBLICATIONS

Guardrail Selection in Line Charts to Contextualize Persuasive Visualizations

Khandaker Abrar Nadib, Marina Kogan, Alexander Lex, and Marina Lisnic.

Computer Graphics Forum (Proceedings of EuroVis 2026), 2026. DOI: 10.1111/cgf.70460

ReVISit 2: A Full Experiment Life Cycle User Study Framework

Zach Cutler, Jack Wilburn, Hilson Shrestha, Yiren Ding, Brian Bollen, **Khandaker Abrar Nadib**, Tingying He, Andrew McNutt, Lane Harrison, and Alexander Lex.

IEEE Transactions on Visualization and Computer Graphics (Proceedings of IEEE VIS 2025, **Best Paper Award**), 2025. DOI: 10.48550/arXiv.2508.03876

Interaction Based Credibility Analysis of News on Facebook Using Machine Learning Methodologies

Sadia Sharmin, Sudipa Saha, Tasin Hoque, and **Khandaker Abrar Nadib**.

In *Proceedings of the 16th International Conference on Signal Image Technology & Internet based Systems (SITIS)*, 2022. DOI: 10.1109/SITIS57111.2022.00077

EDUCATION

University of Utah

Doctor of Philosophy in Computer Science

Salt Lake City, Utah

August 2024 – Present

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

Feb 2017 – May, 2022

RESEARCH EXPERIENCE

Framework for Intervention Against Visualization Recontextualization

2026

Visualization Design Lab & KORE Lab, University of Utah

Salt Lake City, Utah

- Constructed and coded a real-world corpus of social-media chart-and-caption posts spanning visualization design violations and reasoning-based misleaders.
- Currently conducting a design study of misleader interventions and qualitatively analyzing the resulting designs to derive an intervention framework.
- Designing an agentic proof of concept that diagnoses misleading chart-caption pairs and generates interventions through structured visualization specifications.

Guardrail Selection in Line Charts for Persuasive Visualizations

2025

Visualization Design Lab & KORE Lab, University of Utah

Salt Lake City, Utah

- Led a preregistered mixed-design study comparing five contextual guardrail strategies across COVID-19 and stock-market scenarios; recruited 500 participants and analyzed 487 after preregistered exclusions.
- Designed and implemented the interactive study system and analysis pipeline, demonstrating how context-selection strategies differently affect trust, rank-judgment accuracy, and perceived contextual completeness.
- **Technologies:** TypeScript, React, D3.js, Vega-Lite, reVISit, Python, Supabase, and Firebase.

Ranking Visualizations of Correlation (ReVISit Replication Study)

2024

ReVISit 2 Replication Project

University of Utah

- Designed and implemented a replication of Harrison et al.'s correlation JND study using reVISit's dynamic sequencing and staircase designs.
- Conducted a crowdsourced study across multiple visualization conditions (scatterplots, PCPs, hexbins, heatmaps).
- **Status:** Published. Additionally, working on a first-authored short paper.

News Credibility Analysis on Facebook using User Interactions

2021 – 2022

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

- Proposed and evaluated an interaction-based approach for fake news detection on Facebook, using engagement signals rather than language features.
- Trained machine learning models to classify the authenticity of public posts; showed improved performance over content- and NLP-based baselines and language independence.
- **Tech:** Crowdtangle, scikit-learn, pandas, matplotlib.
- **Status:** Published

WORK EXPERIENCE

Graduate Research Assistant

January 2025 – Present

University of Utah, Salt Lake City

Visualization Design Lab, KORE Lab

- Current work supported by a Responsible AI project exploring AI-assisted analysis in sensitive domains, including health-related data.

Graduate Research Fellow

August 2024 – December 2024

University of Utah, Salt Lake City

Visualization Design Lab, KORE Lab

Software Engineer

May 2022 – July 2024

Optimizely, Dhaka

Digital Asset Management (DAM)

November 2022 – July 2024

- Contributed to a microservice-based Digital Asset Management platform by building asset templating, collections, navigation, analytics instrumentation, AI-integrated content generation, database management, and codebase migrations.
- **Technologies:** Python, Flask, JavaScript, TypeScript, React.js, MySQL, MongoDB, Alembic, Celery, Elasticsearch

Asset Renditions (AR)

May 2022 – October 2022

- Designed and maintained a microservice for automated image and video rendition generation, including stateless processing pipelines, asynchronous messaging, and scalable media transformation workflows.
- **Technologies:** Python, FastAPI, MySQL, PostgreSQL, Docker, Kubernetes, Message Queue

PROJECTS

Interactive Guardrail-Integrated Line Chart Platform

2024–2025

- Implemented interactive dashboards used as stimuli in the guardrail selection study, including guardrail sampling logic, randomization.
- **Technologies:** TypeScript, React 18, Vite, Mantine UI, Redux Toolkit, D3.js, Vega/Vega-Lite, Arquero, reVISit, Supabase, Firebase.

Correlation JND Replication Experiments (ReVISit Framework)

2024

- Built a staircase-style experiment to measure just-noticeable differences in correlation across multiple visualization types using reVISit’s dynamic sequencing.
- **Technologies:** TypeScript, React, D3.js, reVISit, Python (pandas, NumPy, SciPy, statsmodels), firebase.

HappyVis: Visualizing Happiness Around the Globe 2024

- Designed a web-based visualization dashboard for the World Happiness Report dataset, enabling exploration of global happiness scores and contributing factors across countries and years.
- Implemented linked views including an interactive choropleth map, trend line charts, and comparative views to analyze relationships between happiness, GDP per capita, life expectancy, and social support.
- **Technologies:** JavaScript, D3.js.

AWARDS AND HONORS

Best Paper Award, IEEE VIS 2025 2025

for “ReVISit 2: A Full Experiment Life Cycle User Study Framework”

Optimizely SPOT Awards (July & October) 2023

Two peer-nominated SPOT awards recognizing problem solving, team contribution, and performance

Board Merit Scholarships (SSC & HSC) 2014, 2016

Education Board Scholarships; ranked 6th (male) in Dhaka Board in HSC

TECHNICAL SKILLS

Research Methods: Online controlled experiments, interviewing, qualitative coding

Analysis: Mixed-effects models, hypothesis testing; Python (pandas, NumPy, SciPy, statsmodels, matplotlib, scikit-learn,)

Visualization & Frontend: D3.js, Vega, Vega-Lite, Altair, React.js, TypeScript, Redux Toolkit

Languages: JS, Python, Java, C/C++, SQL, PL/SQL

Databases: MySQL, Elasticsearch, PostgreSQL, MongoDB, Oracle

Frameworks/Libraries: Flask, FastAPI, Node.js, Bootstrap, Celery, Alembic, Enzyme (JS), React Testing Library

Tools/Software: Git, Docker, Kubernetes, S3, Jupyter, Playwright

Scripting/Markup/Serialization: Bash, TCL, L^AT_EX, YAML, HTML, JSON